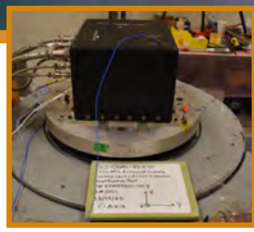
National Aeronautics and
Space Administration

Marshall Space Flight Center

Structural Dynamics Testing

Engineering Solutions for Space Science and Exploration

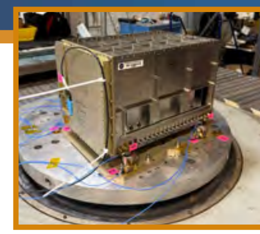
Acoustic acceptance test of NASA plume cover



Vibration qualification test of on-board camera

Modal testing of Artemis launch vehicle at Kennedy Space Center

Separation-shock testing of a panel frangible joint



Vibration qualification test of Space Station locker

Structural Dynamics Testing comprises a wide range of testing techniques to characterize the dynamic behavior of a variety of structures. Dynamic testing may simulate an expected launch environment to ensure the successful performance of a flight article, it may be required for troubleshooting, or it may be conducted to validate structural models used in a variety of analyses. The Structural Dynamics Test Branch has extensive experience in many types of testing, including vibration, acoustic, pyrotechnic shock, fluid slosh, and modal testing. Unique facilities and test equipment combined with the expertise of personnel gained over decades of testing provide a flexible and efficient option for both large and small test articles. Customized fixture design and fabrication can be provided through in-house organizational resources.

Vibration Testing

Vibration testing is conducted utilizing two separate laboratories, each located adjacent to high-bay structures. A total of six electrodynamic exciters are dedicated to development and certification vibration testing of flight and ground support hardware. Controlled excitation up to 65K lbf is provided using multiple digital vibration control systems. These systems provide sine sweep, broad-band random, sine-on random, random-on random, classical shock, and shock response spectrum (SRS) environments. Other control features include 24-bit resolution, real time 64-channel input, tolerance limited spectra, test article response limiting, and real time signal analysis. Shaker head expanders allow mounting surfaces up to 8 ft by 8 ft. Real time acquisition of 250+ time domain response channels that can be formatted or post processed to satisfy conventional or specific test requirements. Extensive experience base includes testing at cryogenic temperatures down to -425F, elevated temperatures, high pressures, or a combination of these environments.

Acoustic Testing

Acoustic testing is conducted in a 5,000 ft^3 reverberation chamber constructed contiguous to an anechoic chamber. This chamber arrangement allows for traditional diffuse field testing, transmission loss testing, or acoustic emission testing. An array of noise sources is available ranging from speakers up to a pair of 150kW pneumatic sources paired to a 29Hz horn yielding 163dB in the diffuse environment. The horn can be configured with a progressive wave tube having 21" x 42" cross-sectional dimensions and varying section lengths.

Test articles up to 500 ft^3 can be tested in the diffuse field of the reverberant chamber. The contiguous 3,000 ft^3 anechoic chamber provides the capability for transmission loss studies of test panels up to 8' x 10' as well as acoustic emission and sound power measurements. Data acquisition of 250+ channels with a broad spectrum of instrumentation and conditioning is available to support these test including microphones, accelerometers, and strain gages. Acoustic tests are controlled based on constant percentage bandwidth feedback with 1/3 octave being typical.

Shock Testing

Pyrotechnic shock testing is conducted in an area equipped for generating dynamic transients with explosive materials. Detonating cord and blasting caps are used to generate flight input transient shock simulation to test hardware commonly mounted on a suspended plate. Shock levels up to 50,000 g's SRS and 10 kHz can be generated. Pneumatic shock testing can generate shock levels up to 100,000 g's and 10 kHz using a tunable beam. Pyrotechnic devices used in aero-space flight applications can be evaluated and characterized as to the SRS response resulting from detonation. Sixteen channels of transient pyrotechnic response data can be acquired real-time and post processed in the time domain or SRS for one pyrotechnic event. Up to 40 additional channels can be conditioned and digitally recorded.

Modal Testing

The MSFC modal test facility is equipped to conduct modal and other related dynamic testing on a variety of hardware, including flight systems, payloads, and components. Testing may be conducted in the MSFC Test facility or at a customer facility. Fixed, free-free, or hybrid boundary conditions may be accommodated. Excitation may be provided by a variety of impulse hammers or electromagnetic or hydraulic shakers using transient, sinusoidal, or random inputs. A broad range of shakers are available from 2-lb to 250-lb output electromagnetic shakers to 1,500-lb output hydraulic shakers. The MSFC modal test facility also has several dynamic signal analyzers providing over 500 channels of simultaneous data acquisition. The modal team also has expertise in non-conventional modal testing and data acquisition techniques, including the development of several slosh test beds and non-contacting, full-field measurement capabilities. These capabilities include laser vibrometry, digital image correlation, and photogrammetry.

Capabilities

East Vibration Laboratory 65klbf shaker

Vibration Testing

- 6 electromagnetic shakers up to 65,000 lbf
- Accelerometers with capability up to 5,000 g's
- Test article accommodation up to 8 ft × 8 ft
- Signal conditioning and data acquisition and control systems with up to 64 channels
- Vibration spectra including: Sine, random, sine on random, random on random, classical shock, and Shock Response Spectrum (SRS)

Anechoic acoustic testing of Space Station locker

Acoustic Testing

- Acoustic environments
  - Diffuse field to 163 dB
  - Progressive wave to 172 dB
- Acoustic Characterization
  - Transmission loss, reverb to anechoic
  - Acoustic emission testing
- Acoustic excitation up to 300 kW acoustic
- Test article accommodation up to 500 ft^3
- Chamber access doors 8 ft × 8 ft

Full-scale payload fairing separation test

Shock Testing

- Pyrotechnics that generate up to 50,000 g's SRS and 10,000 Hz; Pneumatics that generate up to 100,000 g's and 10,000 Hz.
- Accelerometers capable up to 50,000 g's
- Limited thermal conditioning
- 32-channel system for real-time response data
- 11 ft × 9 ft exterior blast room door
- Signal conditioning and digital recording for up to 40 additional high-speed channels
- Time and shock response spectrum (SRS) analysis

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Modal Testing

- Excitation Sources
  - Instrumented hammers from 50 lbf to 50 klbf
  - Electromagnetic shakers from 2 lbf to 250 lbf
  - Hydraulic shakers up to 1,500 lbf
- Large inventory of accelerometers (DC – 20 kHz)
- Signal conditioning and data acquisition over 500 channels simultaneously
- Mobile, readily available equipment for testing at customer facilities

Phantom high-speed video

Non-contacting Optical Measurement Capability

- Photogrammetry
  - 3D dot/speckle pattern tracking
  - 3D full-field motion/displacement measurement
  - 3D transient strain measurement

Trifilar suspension for mass properties measurements

Mass Properties Measurements

- Center of gravity (CG) & moment of inertia (MOI)
  - Custom fit for articles up to 8.5 ft diameter
  - Mass measurement range 300 to 11,000 lbm
  - MOI measurement accuracy ≤ ± 1%
  - CG measurement accuracy ≤ ± 0.1 in

Slosh baffle testing of a cylindrical tank

Fluid Slosh Testing

- Numerous vertical and lateral test beds with embedded transducers.
- Slosh parameter measurement and estimation
- Excitation up to 3000 lbf using hydraulic shakers
- Custom tank configurations using cryogenic liquids or water

Key Benefits

- Experienced team dedicated to structural dynamic testing
- Existing labs and test equipment, along with in-house design and fabrication capability, allow for rapid test planning and build-up
- Broad range of expertise in all facets of structural dynamic testing to support hardware development and qualification through their environmental test campaigns
- Mobile capabilities for testing at customer facilities

Vibration test of a protoflight satellite

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Modal Test of Orbiter Boom Sensor System